DISCONNECTEDNESS DESCENDS ALONG A MORPHISM FROM A GEOMETRICALLY CONNECTED SCHEME OVER A GALOIS EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains the sentence 'Verification omitted.' at the compatibility of the U_W under inclusions of quasi-compact opens; the sentence is reproduced verbatim. The omitted check is a bounded monotonicity statement and the glueing conclusion is otherwise complete as written.

Editorial note. The counted spans carry 1 navigation stub and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-tricky also occurs in chow.tex, but with a completely different statement about morphisms of (2,1)-periodic complexes over a local ring; only the label coincides, so this card counts its own proposition.

Allowed standard prerequisites (editorial).

- Descent of Galois invariant closed subsets. Same chapter, Lemma lemma-closed-fixed-by-Galois, invoked at proof lines 1050-1052. Supplies that the invariant open and closed pieces of W_{k'} come from subsets of W.
- The pieces of a quasi-compact open over the closure are defined over a finite subextension. Same chapter, the Galois action lemma applied to the quasi-compact open W, used at proof lines 1037-1041. Supplies the finite Galois subextension k' and the descent of the two pieces to W_{k'}.
- Continuous image of a connected set is connected. Standard topology, used at proof lines 1031-1034. Supplies that T over the closure lands inside one of the two pieces.

Source reference. Original lemma 038B: sources/stacks-varieties/pinned-original-source.tex, source lines 899–899. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let $\overline{k}/k$ be a (possibly infinite) Galois extension. Let $f : T \to X$ be a morphism of schemes over k. Assume $T_{\overline{k}}$ connected and $X_{\overline{k}}$ disconnected. Then X is disconnected.*

Proof. Write $X_{\overline{k}} = \overline{U} \amalg \overline{V}$ with $\overline{U}$ and $\overline{V}$ open and closed. Denote $\overline{f} : T_{\overline{k}} \to X_{\overline{k}}$ the base change of f. Since $T_{\overline{k}}$ is connected we see that $T_{\overline{k}}$ is contained in either $\overline{f}^{-1}(\overline{U})$ or $\overline{f}^{-1}(\overline{V})$. Say $T_{\overline{k}} \subset \overline{f}^{-1}(\overline{U})$.

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Fix a quasi-compact open $W \subset X$. There exists a finite Galois subextension $\overline{k}/k'/k$ such that $\overline{U} \cap W_{\overline{k}}$ and $\overline{V} \cap W_{\overline{k}}$ come from quasi-compact opens $U', V' \subset W_{k'}$. Then also $W_{k'} = U' \amalg V'$. Consider

$$U'' = \bigcap\nolimits_{\sigma \in \mathrm{Gal}(k'/k)} \sigma(U'), \quad V'' = \bigcup\nolimits_{\sigma \in \mathrm{Gal}(k'/k)} \sigma(V').$$

These are Galois invariant, open and closed, and $W_{k'} = U'' \amalg V''$. By Lemma 038B we get open and closed subsets $U_W, V_W \subset W$ such that $U'' = (U_W)_{k'}$, $V'' = (V_W)_{k'}$ and $W = U_W \amalg V_W$.

We claim that if $W \subset W' \subset X$ are quasi-compact open, then $W \cap U_{W'} = U_W$ and $W \cap V_{W'} = V_W$. Verification omitted. Hence we see that upon defining $U = \bigcup_{W \subset X} U_W$ and $V = \bigcup_{W \subset X} V_W$ we obtain $X = U \amalg V$. It is clear that V is nonempty as it is constructed by taking unions (locally). On the other hand, U is nonempty since it contains $f(T)$ by construction. $\square$